Nikolai Slavov

Contact Information

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ORCID ID: 0000-0003-2035-1820

Education

Princeton University, Princeton, USA

- **Ph.D.** in Molecular and Quantitative Cell Biology — **06.2010**
- **M.A.** in Molecular Biology — **02.2006**
- **Dissertation:** Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations
- **Mentor:** David Botstein

Massachusetts Institute of Technology (MIT), Cambridge, USA

- **B.S.** in Biology — **06.2004**

Academic Appointments

Allen Institute, The Paul G. Allen Frontiers Group

- Allen Distinguished Investigator — **2020–present**

Northeastern University, Boston, USA

- Associate Professor, Department of Bioengineering — **2021–present**
- Faculty Fellow, Barnett Institute — **2017–present**
- Assistant Professor, Department of Bioengineering — **2015–2021**

Harvard University, Cambridge, USA

- Postdoc in the Departments of Systems Biology and Statistics — **2014–2015**

Massachusetts Institute of Technology (MIT), Cambridge, USA

- Postdoc in the van Oudenaarden laboratory, Departments of Biology and Physics — **2011–2013**

Awards & Fellowships

- ✓ Allen Distinguished Investigator ($ 1,500,000) — **2020**
- ✓ Excellence in Mentoring Award — **2018**
- ✓ NIH Director's New Innovator Award ($ 2,352,750) — **2016**
- ✓ Broad Institute of Harvard and MIT SPARC Award — **2014**
- ✓ Princeton University Dean's Award — **2010**
- ✓ IRCSET Postgraduate Research Fellowship (72,000 €) — **2007**
- ✓ Finalist in the Young European Entrepreneur Competition — **2006**
- ✓ Princeton Graduate Fellowship ($ 45,000) — **2004**
- ✓ MIT Undergraduate Fellowship ($ 125,000) — **2001**
- ✓ Eureka Fellowship for Academic Excellence (10,000 €) — **2000**
- ✓ Bronze Medal in the 31^{st} International Chemistry Olympiad (*IChO*) — **1999**
- ✓ National Diploma for Exceptional Achievements in Chemistry — **1999**

SELECTED PEER REVIEWED PUBLICATIONS

[OA] – Open Access article ✉ – Corresponding author Google Scholar Profile

2021 **2021**

✓ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**✉
Multiplexed single-cell proteomics using SCoPE2
Nature Protocols (in press), DOI: 10.1101/2021.03.12.435034 **[OA]**

✓ **Slavov N**✉
Measuring Protein Shapes in Living Cellst
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.1c00376

✓ **Slavov N**✉
Increasing proteomics throughput
Nature Biotechnology, DOI: 10.1038/s41587-021-00881-z

✓ **Slavov N**✉ *et al.*
Voices of biotech research: Single-cell proteomics
Nature Biotechnology, 39, 281286, DOI: 10.1038/s41587-021-00847-1

✓ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, Serra M, Kharchenko P, Koller A & **Slavov N**✉
Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2
Genome Biology, 22, 50 DOI: 10.1186/s13059-021-02267-5 **[OA]**

2020 **2020**

✓ Specht H, **Slavov N**✉
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.0c00675

✓ Petelski AA **Slavov N**✉
Analyzing ribosome remodeling in health and disease
Proteomics, 20, (17) :2000039 DOI: 10.1002/pmic.202000039

✓ **Slavov N**✉
Single-cell protein analysis by mass-spectrometry
Current Opinion in Chemical Biology, 60, 1 - 9 DOI: 10.1016/j.cbpa.2020.04.018

✓ **Slavov N**✉
Unpicking the proteome in single cells
Science, 367 (6477) :512 - 513, DOI: 10.1126/science.aaz6695

2019 **2019**

✓ **Slavov N**✉ *et al.*
Voices in methods development: Single-cell proteomics
Nature Methods, 16, 945 - 951, DOI: 10.1038/s41592-019-0585-6

✓ Huffman RG, Chen AT, Specht H & **Slavov N**✉
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
Journal of Proteome Research, 18, 6, 2493 - 2500, DOI: 10.1021/acs.jproteome.9b00039
↪ Technology feature highlight by *Nature Methods*

✓ Chen AT, Franks A & **Slavov N**✉
DART-ID increases single-cell proteome coverage
PLoS Computational Biology, 5(7):e1007082 DOI: 10.1371/journal.pcbi.1007082 **[OA]**
↪ The single cell proteomics revolution, editorial highlight by *Bioanalysis Zone*

✓ Emmott EP, Jovanovic M & **Slavov N**✉
Approaches for studying ribosome specialization
Trends in Biochemical Sciences, 44(5):478 - 479, DOI: 10.1016/j.tibs.2019.01.008

✓ Emmott EP, Jovanovic M & **Slavov N**✉
Ribosome stoichiometry: from form to function
Trends in Biochemical Sciences, 44(2):95 - 109, DOI: 10.1016/j.tibs.2018.10.009

✓ Malioutov D., Chen T., Jaffe J., Airoldi E., Budnik B. & **Slavov N**✉
Quantifying homologous proteins and proteoforms
Molecular & Cellular Proteomics, 18(1):162 - 168, DOI: 10.1074/mcp.TIR118.000947

2018 **2018**

✓ Specht H. & **Slavov N**✉
Transformative opportunities for single cell proteomics
Journal of Proteome Research, 17 (8), 2565 - 2571 , DOI: 10.1021/acs.jproteome.8b00257
↪ Innovations in Proteomics: The Drive to Single Cells, editorial highlight
↪ Through the Looking Glass of Single Cell Proteomics, highlight by *Technology Networks*

✓ Budnik B., Levy E., Harmange G. & **Slavov N**✉
SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, 19(1):161, DOI: 10.1186/s13059-018-1547-5 **[OA]**
↪ SCoPE-MS – We can finally do single cell proteomics!!!, highlight by *Proteomics News*
↪ Researchers Apply Mass Spec to Single-Cell Proteomics, highlight by *GenomeWeb*
↪ Single-cell proteomics highlight by *the NIH Director's office*
↪ Technology feature highlight by *Nature Methods*
↪ The single cell proteomics revolution, editorial highlight by *Bioanalysis Zone*

✓ Levy E. & **Slavov N**✉
Single cell protein analysis for systems biology
Essays in Biochemistry, 62(4):595 - 605, DOI: 10.1042/EBC20180014

2017 **2017**

✓ Franks A., Airoldi E.M., **Slavov N**✉
Post-transcriptional regulation across human tissues
PLoS Computational Biology, 13(5): e1005535, DOI: 110.1371/journal.pcbi.1005535 **[OA]**
↪ How Statistics Weakened mRNAs Predictive Power highlight by *The Scientist*

✓ Saleh D, Najjar M, Zelic M, Shah S, Nogusa S, Polykratis A, Paczosa MK, Gough PJ, Bertin J, Whalen M, Fitzgerald KA, **Slavov N**, Pasparakis M, Balachandran S, Kelliher M, Mecsas J, Degterev A.
Kinase Activities of RIPK1 and RIPK3 Can Direct IFN-β Synthesis Induced by Lipopolysaccharide.
The Journal of Immunology, 198(11):4435-4447, DOI: 10.4049/jimmunol.1601717

2016 **2016**

✓ Klionsky, *et al.*, **Slavov N**, *et al.*
Guidelines for the use and interpretation of assays for monitoring autophagy
Autophagy, vol. 12, issue 1, 1–222, 12(1):1-222, DOI: 10.1080/15548627.2015.1100356

✓ Di Luca A, Hamill RM, Mullen AM, **Slavov N**, Elia G.
Comparative Proteomic Profiling of Divergent Phenotypes for Water Holding Capacity across the Post Mortem Ageing Period in Porcine Muscle Exudate
PLoS One, 11(3):e0150605, DOI: 10.1371/journal.pone.0150605 **[OA]**

2015 2015

✓ **Slavov N**✉
Making the most of peer review
eLife, 4:e12708, DOI: 10.7554/eLife.12708 **[OA]**

✓ **Slavov N**✉, Semrau S., Airoldi E.M., Budnik B., van Oudenaarden A.
Differential stoichiometry among core ribosomal proteins
Cell Reports, vol. 13, issue 5, 865–873, DOI: 10.1016/j.celrep.2015.09.056 **[OA]**
↪ All Ribosomes Are Created Equal. Really?, highlight by *Trends in Biochemical Sciences*
↪ New direction for tissue engineering and cancer therapeutics, highlight by *Science X*

✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N**, *et al.*
De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
Cell Metabolism, vol. 21, issue 5, 764–776, DOI: 10.1016/j.cmet.2015.04.003

Started Slavov Lab at Northeastern University, August 2015

2014 2014

✓ Malioutov D.✉ , **Slavov N**✉
Convex Total Least Squares
Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 **[OA]**

✓ **Slavov N**✉, Budnik B., Schwab D., Airoldi E., van Oudenaarden A.✉
Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
Cell Reports vol. 7, issue 3, 705–714 **[OA]**

2013 2013

✓ **Slavov N**✉, Carey, J., Linse, S.✉
Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
ACS Chemical Neuroscience, vol. 4, 601–612 **[OA]**

✓ **Slavov N**✉, Botstein D.✉
Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Deregulation of Cell Size and Gene Expression
Molecular Biology of the Cell, vol. 24, issue 2, 157–168 **[OA]**

2012 2012

✓ **Slavov N**✉, van Oudenaarden A.✉
How to Regulate a Gene: To Repress or to Activate?
Molecular Cell, vol. 46, issue 5, 551–552 **[OA]**

✓ **Slavov N**✉, Airoldi E.M., van Oudenaarden A., Botstein D.✉
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 **[OA]**

2011 2011

✓ **Slavov N**✉, Macinskas J., Caudy A., Botstein D.✉
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 **[OA]**

✓ **Slavov N**✉, Botstein D.✉
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 **[OA]**

2010 2010

✓ **Slavov N**✉
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 **[OA]**

✓ **Slavov N**✉, Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University **[OA]**

✓ Silverman SJ., **Slavov N**, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.✉
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 **[OA]**

2009 and before 2009 and before

✓ **Slavov N**✉, Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079-4084 ("In This Issue" article) **[OA]**

✓ Tagkopoulos I.✉ , **Slavov N**✉, Kung S.✉ (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5^{th} IEEE Symposium on Bioinformatics and Bioengineering (BIBE'05).

PATENTS

- **Slavov N**✉, Budnik B., Specht H., Levy E. Mass spectrometry technique for single cell proteomics, 16/251,039 (2019)

PUBLISHED PREPRINTS
GOOGLE SCHOLAR

▷ Leduc A, Huffman RG, **Slavov N**✉ (2021)
Droplet sample preparation for single-cell proteomics applied to the cell cycle
bioRxiv, DOI: 10.1101/2021.04.24.441211 **[OA]**

▷ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**✉ (2021)
Multiplexed single-cell proteomics using SCoPE2
bioRxiv, DOI: 10.1101/2021.03.12.435034 **[OA]**

▷ Specht H, **Slavov N**✉ (2020)
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
bioRxiv, DOI: 10.1101/2020.08.24.264994 **[OA]**

▷ Petelski AA, **Slavov N**✉ (2020)
Analyzing ribosome remodeling in health and disease
arXiv, arXiv:2007.05839 **[OA]**

▷ **Slavov N**✉ (2020)
Single-cell protein analysis by mass-spectrometry
arXiv, arXiv:2004.02069 **[OA]**

▷ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, *et al.* & **Slavov N**✉ (2019)
Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity *bioRxiv*, DOI: 10.1101/665307 **[OA]**
↪ Over 1,000 SINGLE CELL PROTEOMES!, highlight by *Proteomics News*
↪ Technology feature highlight by *Nature Methods*

▷ Specht H, Emmott E, Koller A, **Slavov N**✉ (2019)
High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
bioRxiv, DOI: 10.1101/665307 **[OA]**

▷ Huffman G, Specht H, Chen AT, **Slavov N**✉ (2019)
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
bioRxiv, DOI: 10.1101/512152 **[OA]**

▷ Specht H, Harmange G, Perlman DH, Emmott E, Niziolek Z, Budnik B, **Slavov N**✉ (2018)
Automated sample preparation for high-throughput single-cell proteomics
bioRxiv, DOI: 10.1101/399774 **[OA]**

▷ Chen A, Franks A, **Slavov N**✉ (2018)
DART-ID increases single-cell proteome coverage
bioRxiv, DOI: 10.1101/399121 **[OA]**

▷ Levy E. & **Slavov N**✉ (2018)
Single cell protein analysis for systems biology
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26965v1 **[OA]**

▷ Emmott EP, Jovanovic M & **Slavov N**✉ (2018)
Ribosome stoichiometry: from form to function
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26991v1 **[OA]**

▷ Specht H. & **Slavov N**✉ (2018)
Routinely quantifying single cell proteomes: A new age in quantitative biology and medicine
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26821v1 **[OA]**

▷ Malioutov D., Chen T., Jaffe J., Airoldi E., Carr S., Budnik B., **Slavov N**✉ (2017)
Quantifying homologous proteins and proteoforms
bioRxiv, DOI: 10.1101/168765 **[OA]**

▷ Berg P., Budnik B., **Slavov N**✉, Semrau S. (2017)
Dynamic post-transcriptional regulation during embryonic stem cell differentiation
bioRxiv, DOI: 10.1101/123497 **[OA]**

▷ Budnik B., Levy E., **Slavov N**✉ (2017)
Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
bioRxiv, DOI: 10.1101/102681 **[OA]**
↪ SCoPE-MS – We can finally do single cell proteomics!!!, highlight by *Proteomics News*
↪ Researchers Apply Mass Spec to Single-Cell Proteomics, highlight by *GenomeWeb*
↪ Interview, highlight by *Front Line Genomics*

▷ **Slavov N**✉ (2015)
From differential transcription of ribosomal proteins to differential structure of ribosomes
PeerJ Preprint, DOI: 10.7287/peerj.preprints.1504v1 **[OA]**

▷ Franks A., Airoldi E.M., **Slavov N**✉ (2015)
Post-transcriptional regulation across human tissues
bioRxiv, DOI: 10.1101/020206 **[OA]**

- ▷ **Slavov N**✉, Botstein, D., Caudy A. (2014)
  Extensive regulation of metabolism and growth during the cell division cycle
  bioRxiv, DOI: 10.1101/005629 **[OA]**

- ▷ **Slavov N**✉, Semrau S., Airoldi E.M., Budnik B., van Oudenaarden A. (2014)
  Differential stoichiometry among core ribosomal proteins
  bioRxiv, DOI: 10.1101/005553 **[OA]**

Opinions & Popular Science Publications

- ▷ **Slavov N**✉ (2018) We Must Demand Evidence of Peer Review
  The Scientist, Link

- ▷ **Slavov N**✉ (2017) What was rate limiting was the idea, not the technology
  Front Line Genomics, Link

- ▷ **Slavov N**✉ (2015) Making the most of peer review
  eLife, Link

- ▷ **Slavov N**✉ (2014) Accomplishments Over Accolades!
  The Scientist, Link

- ▷ **Slavov N**✉ (2014) Discoveries lie hidden behind the façade of popular assumptions
  Cell Reporter, Link

- ▷ **Slavov N**✉ (2013) The Best Projects Are Least Obvious
  The Scientist, Link

- ▷ **Slavov N**✉ (2013) A Wonderful Christmas with Mysterious Oscillations
  PubChase, Link

- ▷ **Slavov N**✉ (2012) The Mission of MIT
  The MIT Tech, Link

Invited Keynote Talks

1. Increasing the sensitivity, reliability, reproducibility and throughput of single-cell proteomics
   HUPO Reconnect 2021 World Congress, Human Proteome Organization **Nov.15.2021**
2. Nano proteomic sample preparation for single-cell protein analysis
   International Symposium on Microscale Separations and Bioanalysis **Jul.12.2021**
3. Droplet sample preparation for single-cell proteomics applied to the cell cycle
   2021 AOHUPO, Human Proteome Organization **Jun.30.2021**
4. Computational opportunities for single-cell proteomics
   MaxQuant Summer School **Jun.25.2021**
5. Mapping the transcriptome and proteome of tissues in 3D at single-cell resolution
   Plant Cell Atlas **Apr.08.2021**

6. Characterizing transcriptional and post-transcriptional regulation from single-cell variability
CELLINK Partnership Conference **Dec.11.2020**
7. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity (link)
The 2020 HUPO World Congress, Human Proteome Organization **Oct.20.2020**
8. Single-cell proteomics by mass-spectrometry (link)
Netherlands Proteomics Platform, Utrecht, Netherlands **Jan.17.2020**
9. Quantifying proteins in single cells at high-throughput
Single-cell Omics Symposium at Peking University **Oct.20.2019**
10. High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
European Single Cell Proteomics Conference, Institute of Molecular Pathology (IMP) **Aug.27.2019**
11. Quantifying proteins in single cells at high-throughput
The 28th International KOGO Annual Conference, South Korea **Sept.06.2019**
12. DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
MaxQuant Summer School, University of Wisconsin– Madison, Video **Jul.25.2019**

INVITED DEPARTMENTAL TALKS & VISITS

1. Single-cell proteomic and transcriptomic analysis of macrophage polarization
Department of Cell Biology, Duke University **Jan.14.2021**
2. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Peking University **Oct.11.2020**
3. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Department of Biochemistry, Boston University **Sept.14.2020**
4. Quantifying proteins in single cells at high-throughput
Center for Proteomics and Metabolomics, Leiden University **Jan.16.2020**
5. Quantifying proteins in single cells at high-throughput
Biotech Research and Innovation Center, University of Copenhagen **Jan.15.2020**
6. Coordination among metabolism and cell division
Boston Children Hospital & Harvard Medical School **Nov.14.2019**
7. Single-cell proteomics quantifies the emergence of macrophage heterogeneity
Department of Physiology, Johns Hopkins University School of Medicine **Nov.6.2019**
8. Understanding biological systems: In search of direct causal mechanisms (link) Video
Models, Inference & Algorithms Initiative (MIA), Broad Institute of MIT & Harvard **Apr.3.2019**
9. Understanding and controlling cell differentiation using single-cell mass-spectrometry (link)
Department of Systems Biology (Theory Lunch), Harvard Medical School **May.18.2018**
10. Principles of ribosome-mediated translational regulation
The Center for Engineering in Medicine, Massachusetts General Hospital **Mar.17.2017**
11. Principles of ribosome-mediated translational regulation
Systems Biology Seminar, Boston University **Oct.13.2016**
12. Ribosome-mediated translational regulation: Direct methods for quantifying the ribosome code
Center for Systems Biology, Columbia University **Apr.20.2016**
13. Coordinating metabolism and ribosome-mediated translational regulation
Department of Biology, New York University **Apr.19.2016**

14. Quantifying protein isoforms (link)
Models, Inference & Algorithms Initiative (MIA), Broad Institute **Sept.14.2015**

Invited Presentations for Funding Recommendations

- Current state of single-cell proteomics, standards, potentials and quantitative benchmarks
Functional Proteomics, HubMap & Common Fund, NIH **Oct.8–9.2020**
- Single-cell proteomics
Helmsley charitable trust, Gut cell atlas meeting, NYC **Oct.28.2019**
- Transformative opportunities for single-cell proteomics
NHLBI Single-cell workshop, National Institutes of Health (NIH) **Oct.17.2019**

Invited Conference Talks

1. Droplet sample preparation for single-cell proteomics applied to the cell cycle
CZI Seed Network Webinar **Jul.20.2020**
2. Single-cell Proteomic and Transcriptomic Analysis of Macrophage Heterogeneity (link)
HUPO World Congress **Oct.20.2020**
3. Single-cell protein analysis by mass-spectrometry (link)
The future of single cell analysis, Earlham Institute **Feb.16.2020**
4. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity (link)
Applied Bioinformatics in Life Sciences **Feb.14.2020**
5. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity (link)
Cellular and Molecular Bioengineering Conference, BMES **Jan.3.2020**
6. Single-cell proteomics for studying regeneration and aging (link)
MDI Biological Symposium for Regeneration and Aging **Oct.9.2019**
7. Mapping the Transcriptome and Proteome of Human Testis in 3D
CZI Human Cell Atlas Seed Networks Meeting **Jul.30.2019**
8. Transformative opportunities for single cell proteomics
The Greater Boston Mass Spectrometry Discussion Group **Dec.13.2018**
9. Understanding and controlling cell differentiation based on comprehensive mass-spec quantification of proteins and signaling in single cells
UPENN Single Cell Biology Symposium, Video **Nov.13.2018**
10. Chair for the Single Cell Proteomics session of HUPO
Automated sample preparation for high-throughput single-cell proteomics **Oct.3.2018**
11. Using single-cell proteomics to engineer directed cell differentiation
Bioengineering & Translational Medicine Conference **Sept.27.2018**
12. Analyzing single cell proteomics data
ETH, Biognosys Discovery Proteomics Seminar **Jul.7.2018**
13. Mass–spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
CSHL Meeting: Single Cell Analyses **Nov.10.2017**

14. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Annual Single Cell Analysis USA Congress, Boston **Oct.23.2017**
15. Quantifying protein heterogeneity during the differentiation of single stem cells
Global CESI-MS Symposium **Oct.06.2017**
16. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Festival of Genomics, Boston **Oct.04.2017**
17. From differential transcription of ribosomal proteins to differential structure of ribosomes
Genetics, Genomics and Beyond, Calico **Sept.16.2017**
18. Differential stoichiometry among core ribosomal proteins
CSHL Translational Control, Cold Spring Harbor Laboratory **Sept.08.2016**
19. Principles of ribosome-mediated translational regulation
Center for Interdisciplinary Research on Complex Systems **Apr.12.2016**
20. Coordination among cell growth, cell division and gene expression
Broad Institute of MIT and Harvard, YSB **Dec.9.2015**
21. Coordinating cell growth and metabolism with ribosome-mediated translational regulation
Northeastern University, Biology Department Colloquium **Oct.19.2015**
22. Direct quantification of highly homologous proteins by mass–spectrometry indicates physiological changes in the protein composition of the ribosome
Broad Institute of MIT and Harvard, CBBO **Jun.25.2014**
23. Variable stoichiometry among core ribosomal proteins
Broad Institute of MIT and Harvard **Jun.19.2014**
24. Quantification of translational regulation of homologous proteins with unprecedented accuracy
Harvard HMS Systems Biology Theory Lunch **Mar.21.2014**
25. Exponentially growing cells can support a constant growth rate by very different metabolic strategies
Fourth Annual Physical Sciences in Oncology Meeting, NCI **Apr.18.2013**
26. Experimentally identified trade-offs of aerobic glycolysis
Center of Cancer Systems Biology at Tufts **Dec.18.2012**
27. Trade-offs of aerobic glycolysis
Harvard HMS Systems Biology Theory Lunch **Sep.12.2012**
28. The dynamics of exponential cell growth at constant growth rate
MIT Physical Sciences-Oncology Center **Jun.11.2012**
29. Metabolic cycling without cell division cycling
Bauer Forum at Harvard FAS Systems Biology **Apr.15.2011**

Conference Talks

1. Quantifying homologous proteins and proteoforms
US HUPO Conference **Mar.14.2016**
2. Variable stoichiometry among core ribosomal proteins
2014 ASCB/IFCB Annual Meeting **Dec.8.2014**
3. The dynamics of exponential growth
GSA Yeast Meeting, Princeton University **Aug.5.2012**

4. The dynamics of exponential growth
   Broad Institute of Harvard and MIT **Apr.17.2012**
5. Inference of Sparse Networks with Unobserved Variables. Application to Gene Regulatory Networks
   International Conference on Artificial Intelligence and Statistics **May.14.2010**

CURRENT EXTERNAL FUNDING (GRANTS)

- ✓ NIH Director's New Innovator Award ($ 2, 352, 750) **2016-2021**
  Sole PI, *100 %* DP2GM123497, Ribosome-mediated translational regulation during stem cell differentiation
- ✓ Allen Distinguished Investigator Award ($ 1, 500, 000) **2020-2023**
  Sole PI, *92 %*, Tracking proteome dynamics in single cells
- ✓ CZI Seed Networks ($ 400, 000) **2019-2022**
  Lead PI, *40 %*, Mapping the Transcriptome and Proteome of Human Testis in 3D
- ✓ Merck Exploratory Science Center Fellowship (Approved, not awarded yet: $ 341, 177) **2020-2021**
  Sole PI, *100 %*, Characterizing the protein networks underpinning macrophage polarization
- ✓ NIH/ NHGRI **2020-2025**
  Co-investigator, R01HG011087, Single-cell direct RNA sequencing using electrical zero-mode waveguides and engineered reverse transcriptases
- ✓ NIH/NCI R21 **2020-2022**
  Co-investigator, R21CA246150, Discovering proteins that explain single-cell heterogeneity in fibrillar migration

PENDING EXTERNAL FUNDING (GRANTS)

COMPLETED FUNDING

- ✓ Merck Exploratory Science Center Fellowship ($ 242, 827) **2019-2020**
  PI, *100 %*, Developing next generation single-cell proteomics and using it to characterize the functional polarization of macrophages
- ✓ Sanofi iAward ($ 196, 000) **2018**
  PI, *100 %*, Developing a technology platform for discovering biomarkers and drug targets
- ✓ NEU Tier I Award ($ 50, 000) **2016**
  PI, *50 %*, **Internal funding**
- ✓ Broad Institute of Harvard and MIT SPARC Award ($ 100, 000) **2014**
  PI, *100 %*
- ✓ IRCSET Postgraduate Research Fellowship (72, 000 €) **2007**
  PI, *100 %*

TEACHING & ADVISING

Courses Taught

Teaching websites

Methods of Bioengineering: Mass-spectrometry analysis, Northeastern University **2020 – 2020**
Introduction to mass-spectrometry analysis, Website with lectures

Principles of Bioengineering, Northeastern University **2020 – 2020**
Helping introduce doctoral students to research, Website

Method and Logic in Systems Biology and Bioengineering, Northeastern University **2019 – 2019**
Discussion of seminal papers, Website

Mathematical Methods for Engineers, Northeastern University **2016 – 2020**
Introduction to major concepts and methods in linear algebra differential equations, Website

Method and Logic in Quantitative Biology, Princeton University **2009 – 2010**
Introduction to major ideas and concepts in quantitative biology based on primary literature

Residential Graduate Student at Forbes College, Princeton University **2009 – 2010**
Advised students; organized language tables, sport events and trips

Integrated Quantitative Introduction to Natural Sciences, Princeton University **2006 – 2010**
Interdisciplinary curriculum including physics, computer science, biochemistry, genetics, physiology and emphasizing quantitative problem solving and the connections among these disciplines

Terrascope, MIT **2002 – 2003**
Selected for two subsequent years to lead undergraduate student groups working on complex real-world problems

Biochemistry, MIT **2003 – 2004**
Taught precepts and helped undergraduate students prepare for examinations

Research associates at Northeastern University — Alumni website

Dr. Edward Emmott — *Currently Tenure-track professor at Liverpool University*
Dr. Toni Koller — *Currently Director of Koch Proteomics Facility at MIT*
Dr. David Perlman — *Currently Senior Principal Scientist at Merck*

Graduate Students at Northeastern University — Group members website

Harrison Specht, Bioengineering PhD student — *expected graduation: 2021*
Aleksandra Petelski, Bioengineering PhD student — *expected graduation: 2022*
R. Gray Huffman, Bioengineering PhD student — *expected graduation: 2022*
Jason Derks, Bioengineering PhD student — *expected graduation: 2022*
Tianchi Chen, Bioengineering PhD student (currently in Sontag group) — *expected graduation: 2022*
Andrew Leduc, Bioengineering PhD student — *expected graduation: 2023*
Shiri Tsour, Bioengineering PhD student — *expected graduation: 2023*
Joseph Collura, Bioengineering MS student — *expected graduation: 2022*
Collin Gwilt, Bioengineering MS student — *expected graduation: 2022*

Undergraduate students at Northeastern University — Alumni website

Ezra Levy	*Currently PhD student at Princeton University*
Guillaume Harmange	*Currently PhD student at University of Pennsylvania*
Erin Provost	*Currently Research Associate I at Atalanta Therapeutics*
Albert Chen	*Currently Research Associate the Broad Institute*
Kaely Gallagher	*Currently Engineer at Sana Biotechnology, Inc.*
Ryan Posey	*Currently Research Associate at Beth Israel Deaconess Medical Center*
Aanie Phillips	*Currently Medical student at Baylor College of Medicine*
Matthew Dowling	*Currently Northeastern MS student*
Levon Rodrguez	*Currently Northeastern undergraduate student*
Sebastian Hymson	*Currently Northeastern undergraduate student*

Service: Scientific Leadership

- Organizing the fourth Single-Cell Proteomics Conference (SCP2021) — **Aug 16–18 2021**
- Member of a think tank on Functional Proteomics NIH (OD) — **Sept 8–9 2020**
- Organized the third Single-Cell Proteomics Conference (SCP2020) — **Aug 18–19 2020**
- Co-organized Learning Meaningful Representations of Life workshop at NeurIPS — **Dec 13 2019**
- Member of a think-tank on single-cell analysis NIH (NHLBI) — **Oct 17–18 2019**
- Organized the second Single-Cell Proteomics Conference (SCP2019) — **June 10–12 2019**
- Chair for the fifth Annual Single Cell Analysis USA Congress — **May 14 2019**
- Chair for the Single Cell Proteomics session, HUPO — **Oct 1–3 2018**
- Organized the first Single-Cell Proteomics Conference (SCP2018) — **June 9–10 2018**

Service: Professional Activities

- **Peer-review:** Nature, Science, PNAS, Nature Molecular and Structural Biology, Molecular Biology of the Cell, PLoS, Nucleic Acids Research, Bioinformatics, Annals of Applied Statistics, Cell Reports, eLife, Nature Communications || Publons Profile
- **Editorial roles:** Editor for eLife and PeerJ, bioRxiv Affiliate, Member of the editorial board of Journal of Proteome Research

Service: Department Committees

- Member of the Search Committee for Barnett Institute Director — **2018–2020**
- Member of the Departmental Faculty Search Committee — **2016–2020**
- Member of the Graduate Student Admissions Committee — **2017–2020**